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Flash: fast metrology of atomic assembly

Making materials manufacturing faster, cheaper, and more predictable by tracking nucleation

Summary

There is a substantial gap between gram-scale discovery and scalable manufacturing for new and mature materials. The gap arises because lab-scale recipes do not as a rule transfer to a production line but instead take many trials to re-engineer. Manufacturing competitiveness thus relies on the ability to predict large-scale behaviour from rapid iteration at smaller scales. For many materials, we are missing the key piece to this cycle: how materials first assemble, via a process called nucleation. That encodes the key information that transfers across all scales and enables iteration and predictive design, such as with digital twins. Flash will build the measurement tools to track nucleation at manufacturing-relevant conditions, advance them through validation, and connect them to users. These tools will make it possible to deploy new materials faster and help European manufacturers compete internationally in the XXI century.

Context

Materials like iron and silicon have defined technological ages, but each new flagship material required mastery of new materials science to translate into new tools and new production methods. For steel and silicon, these were control of microstructure and impurities, respectively, which yielded furnace, quenching, and cleanroom infrastructures. Today, we are discovering new materials at an unprecedented pace. They are most important where scale matters, for example in computing, transitioning to a carbon-free economy, and capturing carbon. To unlock their potential, we must learn how to make them economically and at scale. For example, for batteries or carbon removal, that means kiloton quantities. The path from discovery to impact lies through unglamorous behind-the-scenes scale-up engineering, where the speed and scale of iteration matter as much as the material itself.

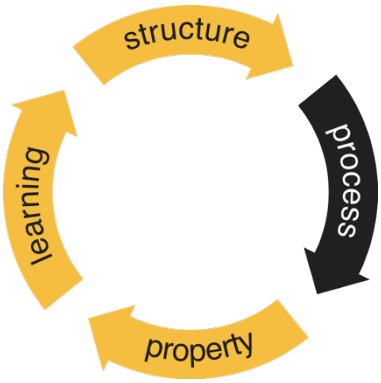


Fig. 1: The cycle of R&D iteration: structure, process, properties, and learning.

As the world becomes more competitive, we face renewed urgency to develop both speed and efficiency for materials manufacturing. The faster we bring new materials into products, the better for both our society, our security, and our economy. Engineering improvements come via a cycle between structure, process, properties, and learning. Each iteration consists of making the material, learning from testing it, and changing the process to improve the next trial. For example, the silicon anode producer Sila Nano has iterated more than 70 000 times over a decade of development. By cost and time, this R&D cycle (Fig. 1) is the bottleneck in bringing new materials to market. The iterations, progressively increasing in cost from grams to tons, are the gap between discovery and use: scale-up. The technical part of 'scale-up' is turning a recipe that makes the first gram of a material into the factory that reliably makes it at tonne scale, consistently, and at the top quality. An engineer in a battery materials company called accelerating scale-up *"the most relevant challenge in industry right now"* since it drives competitiveness. Today, trial and error determine the iteration rate due to post-mortem analyses or proxies being easier than metrology that could directly detect how materials form.

We can no longer afford the wait and expense for trial-and-error deployment, but new understanding and new models could accelerate the cycle. Whenever a physically predictive model became available, the cost and speed of development rose dramatically. For one historical example, when Boeing introduced computer-aided design (CAD) in the 1990s, they cut the number of iterations by 60-90% and cost of development by 15% despite increasing system

complexity, in part by reducing the number of physical mock-up planes to one [1]. CAD has become a foundational tool in developing combustion engines and semiconductor chips; as the precision and scale of these technologies have increased, virtually every iteration of process, chip, or device employs CAD before any physical experiments.

Bottleneck

If predictive models are accelerants, what stands in the way of a CAD for materials manufacturing? The missing piece is the process by which a material assembles from atoms, called nucleation (Appendix C). There are two things to know, control, or exploit about nucleation: its rate and its mechanism; the mechanism includes possible intermediates. Nucleation rates vary exponentially with input concentrations and are the most sensitive parameter in computational fluid dynamics (CFD) and digital-twin simulations of reactors. Measuring and manipulating them is a strong lever on the quality of manufacturing. The absence of this data limits CFD-based simulations or digital twins for processes that involve nucleation; it is the missing piece in modelling manufacturing for materials.

Crucially, the independent variables for nucleation are transferable across manufacturing scales, which enables the data to translate from grams to tons. For example, the rate of nucleation measured in a small flow cell will be the same as in a kiloton-scale reactor for the same local conditions (e.g., temperature, concentrations, Reynolds number). The reactor design and size determine which conditions take place, and nucleation determines how the material then forms.

Our inability to measure nucleation makes the reactor a black box. We have not measured nucleation well because we have lacked the tools for it, mainly in time resolution and throughput. For example, in a 12-hour precipitation of a battery cathode, my work shows the solid forms in ≤ 50 msec (best so far ≥ 5 sec). Even if a measurement of this fast nucleation takes an hour, that could save a further 2 days of ex-situ testing and accelerate iteration by ≥ 50 times.

This singular physical phenomenon separates us from predictive design of materials manufacturing. Characterizing it will open the black box of iteration in manufacturing and enable model-based design of manufacturing. That will in turn reduce iteration time and count in manufacturing R&D. In addition to being integral for process simulation, this new process data will enable first-principles predictions for new materials (Appendix E), CFD-based digital twins, and inline monitoring. Theory of change can be found in Appendix A.

Solution

Flash will build the tools for observing manufacturing-relevant nucleation. Several measurement modalities could provide complementary information on nucleation but need improvement. They lie on a Pareto front between time and size resolution, throughput, and depth of chemical, structural, or statistical information they offer (Appendix D). These include light scattering [2–4], nonlinear optics [5,6], x-ray diffraction [7,8], nuclear magnetic resonance [6], and possibly others. These tools can provide measurements of process rates and scale-free quantities such as nucleation at well-defined process conditions. Some are attempting commercialization for beachhead markets. To turn these candidates into an industry-ready toolkit interoperable across verticals will require pre-commercial development to counterbalance the VC push for immediate if more modest cash flow, and possible further improvement for the specific task of measuring nucleation for manufacturing. To achieve success, these must observe a material transform in production-relevant conditions, sufficiently fast for iteration and production of large-scale data (Appendix D). Flash will build and validate these tools, prepare them for deployment in industry, and ensure the newly available data finds use in model development.

Economic benefit

Developing the metrology toolkit to measure nucleation will accelerate the scale-up of new materials by informing and accelerating iteration and modelling at conditions equivalent to at-scale production. In this way, compressing the timelines between discovery and deployment in products will support globally competitive European manufacturing. The inability to control nanoscale manufacturing costs chemical manufacturers *"hundreds of €M in losses every year,*

but accepting the loss as impossible to improve", via a microscope developer. Importantly, despite standing to benefit from nucleation metrology, manufacturers lack the specialist skills to develop them in-house. While the goal of Flash is to accelerate multiple materials verticals, two key examples below illustrate its €Bn potential and the diversity of possible applications.

First is a zinc-based metal-organic framework (MOF) material, CALF-20 [9]. Its current price, well over €50/kg, is too expensive for its intended application: large-scale carbon capture. However, the cost of its components is under €4/kg, well below the price, about €10/kg, needed to trap point-source CO_2 at market-clearing ≈€100/ton. Closing the gap between current prices and the target is possible but will require improvements in scale and efficiency. The metrology in Flash will accelerate the solution to these manufacturing challenges. If successful, this will advance a diverse class of framework materials, MOFs, that can enable cost-effective carbon capture as well as, e.g., sensing and gas handling, yet specifically struggle with scale-up [10,11].

Second is lithium iron phosphate (LFP), the highest-volume cathode for automotive and grid-storage lithium-ion batteries; cathodes are the dominant cost drivers of all batteries [12,13]. Today, European manufacturing of LFP is not competitive with Chinese producers because of the latter's higher scale and densities, which propagate to the cost and performance of the overall batteries. Improving LFP to compressed densities above 2.6 g/cc ("fourth generation" equivalent) is the only route to competitiveness. Competitive manufacturing is only achievable with control of the particle sizes and nucleation rates during production. This gap to high-quality LFP and all similar cathodes is the engineering of precisely the type enabled by the data that this challenge produces. Solving the process measurement bottleneck could enable competitive European production of these materials. Related are the respective workhorses for sodium-ion batteries: sodium iron pyrophosphate and sodium iron sulfate.

Challenge Design

Flash asks metrology developers to upgrade their instruments towards measurement of nucleation with sufficient throughput, resolution, and versatility for manufacturing and modelling. Participants will build the characterization tools and collect data on key model materials to validate their performance and meet baseline technical benchmarks (stages 1-2). The challenge team and participants will solicit industry feedback to inform more direct and specific industry-informed benchmarks for integration into industry workflows for stage 3.

The challenge will have three stages: the first two stages open (12 months each) and a team-based third stage (12 months). In the first stage, performers will progress towards baseline technical benchmarks on e.g. time resolution and throughput, and on the ability to measure diverse materials and manufacturing routes. In the second, performers will demonstrate their tools on a set of select model materials and produce open-use datasets. Also in the second stage, the program and performers will seek industry partners to inform additional industry-specific metrics the third stage. These may include additional technical or product development benchmarks such as data processing, uptime or longevity, or further information on the nucleation process. The government's role is to approve the fit and select additional criteria for stage 3 based on industry feedback. In stage 3, qualification and testing with industry partners will be facilitated. Material transfer agreements and non-disclosures are allowed if needed.

Performers (all stages) and partners (stage 3) are expected to retain IP and may file patents. The government can retain free-use "government purpose rights". Participants are expected to make their tools or services commercially available, or have a go to market strategy, and either form entities to facilitate that if not already formed, or license to external equipment manufacturers by the end of stage 3. Additional challenge components will jump-start an ecosystem around the data and ensuring that the data to be produced by the instrumentation has wide utility.

Flash focuses on the nucleation of materials; device-level testing and qualification are out of scope. Pharmaceuticals are out of scope because homogeneous small-molecule nucleation tends to be slow, and clinical trials present the larger bottleneck. Subsequent processes, e.g., grain growth, are also often needed, but are out of scope. They are slower, start from the products of nucleation, carry less sensitivity for modelling, and are easier to measure with the same metrology if Flash succeeds. Heterogeneous nucleation is in scope; follow-on CFD is not.

Appendices

Appendix A: Theory of change

The challenge is designed to accelerate the cycles of iteration between a material's discovery and its production at scale. The ability to observe how the material assembles (nucleation, Appendix C) is *sine qua non* for doing so. The challenge's participants will create and deploy the instrumentation to measure it. This will accelerate progress:

1. The instruments themselves can be deployed in industry or used as services. Monitoring the manufacturing process itself accelerates the feedback cycle by eliminating wait for post-mortem characterization.
2. The data from key model materials will provide the necessary parameters for computational chemical engineering of their manufacturing processes, engaging a new community (CFD modellers) who have been limited by the dearth of this data.
3. The same data will provide the ground truth for first-principles predictions (Appendix E) and make an explicit connection to computational chemistry and AI-driven materials discovery for their vertical integration.
4. The instruments augment the integrative capabilities of autonomous laboratories and agentic science with new and rich sources of data.

Every one of the four reduces the number of manufacturing iterations for a material's scale-up. Broadly, this is a model-based strategy: to invent the tools to open the "black box" of a chemical process and create the ability to extrapolate beyond its immediate dataset. In simplest terms, if we learn the rate law for a reaction from short-time measurements, we can predict the products far into the future without re-measuring.

This strategy complements "data-driven" approaches. A data-driven approach continues to treat the reactor as a black box and employs various designs of experiment based on Bayesian optimization, literature analogues, or AI to explore uncertainty in parameter space. The specifics may look different for different companies and verticals, but all encounter a common physical limit: the recipe to make the first gram of a material will not transfer to making a tonne; atoms do not scale like software or AI. For example, a larger reactor made of a different material can throw the process off and require a redesign (a MOF company story). A reactor change meant setbacks and costs for data-driven re-tuning.

Appendix B: Why a challenge? Why Germany?

The challenge format is best for the task at hand: developing new metrology tools, ensuring the unique new data they provide make its way to its users for demonstrating its potential, and connecting the makers of new equipment to funding and users.

There are always multiple ways to measure how a material assembles. Some of these modalities are complementary to one another. For example, x-rays are sensitive to structure and ordering, but miss protons in materials that have them, like porous molecular frameworks. In turn, NMR picks up protons selectively by their spin and distinguishes intermediates by chemical shift. A challenge that funds teams spanning complementary modalities can produce a result that is larger than the sum of its parts. While each method supports an expert community, the skills are largely non-overlapping, and centralizing the effort under the roof of one research organization does not produce significant additional benefit.

Multiple participants can compete within each modality of measurement. At the same time, multi-modal measurement of the same materials with distinct tools (e.g. as above x-rays and NMR on a MOF) helps to validate both sets of results and prepares a more useful dataset for modelling and industrial use. The challenge incorporates this coordination in stage 2 of 3. In the same stage, leveraging existing or public infrastructure such as synchrotron facilities for x-rays is feasible within a challenge format.

Finally, the challenge format and coordination around model materials helps with facilitating connections to the industrial users or buyers of the metrology to be developed in the challenge. Germany possesses both a strong community of measurement science (e.g., synchrotron

facilities such as DESY and BESSY for x-rays), institutes which work on industry-facing science (e.g., Fraunhofer) and a strong industry base focusing on chemical engineering and materials manufacturing, which will benefit from the success of the challenge.

Appendix C: Why does nucleation matter?

Data and metrology are the bottleneck to designing and optimizing syntheses, but nucleation is the reason for that. All materials assemble when atoms rearrange themselves into new ordered patterns. The biggest distinction between syntheses is whether extraneous atoms must escape (e.g. precipitation) or if the substance starts out pure (e.g. freezing from a melt). Almost all manufacturing proceeds via a phenomenon called nucleation, where the smallest discrete units of the final product first emerge. These units do not have to be small; sometimes they can have many thousands of atoms [14]. This is often the case for molecular crystals such as pharmaceuticals.

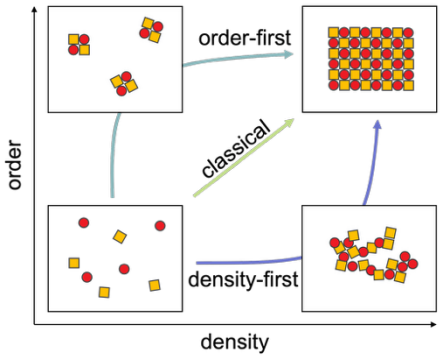


Fig. 2: Three basic kinds of paths for a material to assemble: developing order or final density first, or via classical nucleation, where both are simultaneous.

Manufacturing routes such as freezing a melt, where all atoms are needed, are common to metallurgy and semiconductor manufacturing. By contrast, solution-based syntheses, such as co-precipitation and slow crystallization, some impurities or solvents must escape. These more complex manufacturing routes are key to emerging and mature chemistries across batteries, CO_2 capture, pharmaceuticals, nanomaterials, and cement, among others [2,15–19].

For materials more than for molecules, purity and microstructure determine functionality. The trajectory of a synthesis sets the content of defects and surfaces in the final materials. Predicting the paths of syntheses enables tuning the syntheses to improve quality, transfer processes across reactors, and increase production scale. This is the key bottleneck of scale-up [2,11].

Control of nucleation is extraordinarily important to all of this. If nucleation is too slow, like for many pharmaceuticals, then crystals come out large, our bodies do not have time to dissolve and absorb the drug, and it fails. If it is too fast, the material may fail to order, and its quality will remain suboptimal. This can occur for battery cathodes, where the centers of particles remain of poor quality [20–22]. There are two things to know, control, or exploit about nucleation: its rate and its mechanism. Nucleation rates vary exponentially with input concentrations and are the most sensitive parameter in fluid dynamics and digital twin simulations of a crystallization reactor. Measuring them and manipulating them is a strong lever on the quality of manufacturing.

Knowing the mechanism helps to manipulate the rate. Nucleation proceeds along two general coordinates: density and order [23] (Fig. 2). The classical assumption is that the final phase emerges simultaneously dense and ordered; this is closest to the truth for the simplest materials (e.g. silicon, iron, ice). However, many materials deviate from this, especially at fast process rates relevant to scale-up. For example, gypsum $CaSO_4$ orders first and then agglomerates [8], while others such as silicates, carbonates, or metal-organic frameworks [6,24] proceed via clusters or dense liquids [25], i.e., densifying first, then ordering afterwards.

Appendix D: How fast is fast enough for a measurement of nucleation?

Two metrics matter for a tool to characterize nucleation: time resolution to detect the fastest transformation, and throughput to interface with optimization and engineering in reasonable time (Fig. 3). The target time resolution must be sufficient to resolve the formation of a material on timescales comparable with mixing of the components that form it (chemical engineering jargon: Damköhler number). My own work on battery materials points to such timescales being 50 msec or faster, and that comports with literature effectively giving up on understanding nucleation for battery materials [21]. This makes battery cathodes an appropriately difficult benchmark: solving for them will guarantee applicability for other, slower and simpler chemistries. Furthermore, if the long-term goal is a high manufacturing rate for scaled-up manufacturing, slowing down a process because we cannot control its start is not feasible. Throughput must be sufficiently high

that the measurement does not become its own bottleneck even when screening multiple sets of experimental condition.

Throughput includes flushing the instrument from solids, dead times, and switches between samples. Throughput must be sufficiently high to (1) not create a bottleneck for any given chemistry in industry, and (2) enable the construction of large-scale datasets (100 000 data points) for computational studies in reasonable time, e.g. a year. The first is not possible to estimate without insider information. For the second, a throughput of 10 per hour yields 87600 per year of continuous operation, which we take as benchmark.

These two metrics are the most important ones, so they form the gate for the first stage of the challenge (Appendix F). Additional engineering will be chemistry and modality specific, for example sample delivery will need to tolerate harsh chemistries (e.g. acids or bases) while preserving transparency to measurement; tools can further have constraints based on the minimum detection size (optics), penetration depth (optics), or contrast mechanism.

Beyond a rate, additional information obtainable from measurements helps to pinpoint the mechanism. This can include chemical information on intermediates (from e.g. nuclear magnetic resonance) or probabilistic information from repeated measurements. The latter distinguishes, in simplest terms, whether a solid is detected at half size every time (nucleation complete and growth ongoing) or at full size but only half of the time (incomplete nucleation) along different pathways in Fig. 2. Obtaining and analyzing this probabilistic information for the first time has recently led to updating nucleation theory for solids as simple as noble gases [7,26].

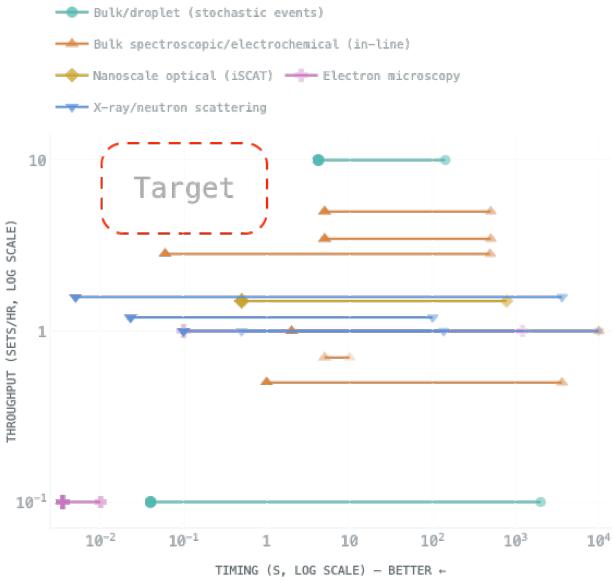


Fig. 3: Time resolution and throughput of measurement methods for chemically initiated nucleation and new material formation. There are no demonstrated methods to beat 3 conditions per hour throughput and 1 second time resolution.

Appendix E: In-situ data enables competitive first-principles prediction

First-principles prediction of nucleation is a grand challenge of materials science. Solving it unlocks computational design of manufacturing processes in silico, prior to making financial investments in large-scale production equipment. This capacity would be transformational for business development. As a necessary disclaimer, no computational models fully eliminate prototyping, but a digital twin of the manufacturing process, of which nucleation is an indispensable component, cuts iteration to the very minimum. Predicting how a material nucleates, like predicting how a protein folds, is a computationally tractable problem. Both yield to straightforward mathematical evaluation and ranking of prediction methods.

Computational chemists have not invented generalizable predictions of nucleation because they have lacked the data for benchmarking theoretical predictions, except for extremely model chemicals such as water, salt, or urea [23,27]. The proposed challenge solves this data bottleneck. Indeed, producing high-quality reference experimental data is a way to showcase the capabilities of the instruments to be developed in the challenge. Data from fast-timescale measurements of nucleation on model materials provides premier ground truth for benchmarking computational predictions in the same way that the Protein Data Bank provided data for the Critical Assessment of Structure Prediction (CASP) competition to predict the folded structures of proteins.

The goal of first-principles prediction is to enable knowing how to make any new material immediately as it is discovered. Nucleation is a harder problem than protein structure because it is kinetic rather than purely thermodynamic. However, the history of CASP lends optimism to a competition to predict how materials may nucleate from first principles: about 97% of the roughly

130 000 experimental structures used to train AlphaFold and AlphaFold2, widely considered to decisively "solve" CASP as a general problem in 2018-2020, were not yet measured when CASP started in 1994. To run a competition to predict nucleation from first principles, we need four things: criteria for evaluation, model systems, resources, and incentives.

- Criteria for evaluation: these are qualitative and quantitative. Qualitative: correctness of nucleation pathway (lowest free energy path) and identity of any intermediates as in Fig. 2. Quantitative: prediction of nucleation rate for a given chemical composition input (concentrations and temperature). These metrics should aim to identify the most useful first-principles simulations for continuum engineering and may need to include predicting additional parameters (e.g. Schmidt ion number).
- Model material systems: since general prediction tools do not currently exist, the complexity of the materials to be predicted should start low and increase over time. The first round of the competition should include e.g. off-patent pharmaceuticals that yield well to inexpensive large-scale simulation to engage the community. Later rounds should introduce additional chemical complexity such as acid-base equilibria (in e.g. MOFs or battery cathodes) which require higher-level theory to simulate accurately. While heterogeneous nucleation (at a surface) can contribute to processing if the overall rate is slow, homogeneous nucleation is expected to dominate at fast manufacturing rates and high driving forces, as for inorganic materials. This makes it possible without loss of utility to focus on homogeneous nucleation in the competition until the field is sufficiently mature and engaged.
- Resources: prediction of nucleation pathways or rates is a problem of both thermodynamics and kinetics, which makes it more expensive than structure prediction. A compute budget, if it can be secured from interested industry, would encourage participation.
- Incentives: again, since nucleation predictions are more complex and (at least currently) more costly than structure predictions, prizes would help to engage and connect the community. These do not need to be large (CASP had none); compute allocations or GPUs would make for good prizes.

Appendix F: challenge design details

Performer evaluation criteria and budget

1. For each characterization mode, either time resolution, or throughput, or both, must improve to be sufficient for measuring transient phase transformations at conditions and speeds relevant to production. The precise shortcomings and required speeds differ by measurement technique. An exemplar set of requirements is as follows:
    a. Interoperability across at least two specified materials verticals and manufacturing methods, e.g., batteries and catalysts, and e.g. precipitation and extrusion.
    b. Ability to measure high concentrations (e.g., ≥1 M active materials, or suspensions, for precipitations, or all-solid materials) of the relevant flowing samples or transforming solids. This is necessary for probing scalable manufacturing processes rather than academic or dilute ones.
    c. Resolution sufficient to observe nucleation and growth. For, e.g., probing precipitation of battery or MOF materials, time resolution faster than 50 msec, with 1 msec as a stretch goal, would be necessary (Appendix D).
    d. Throughput: measurement time below 10 minutes per condition, with 1 minute as a stretch goal (Appendix D).
    e. Deconvolution of statistical information from multi-particle ensembles (optics or x-rays) or chemically specific information (NMR) is strongly desirable. See Appendix C.
    f. Scalable data processing for evaluating kinetics if the measurement produces large quantities of raw data (e.g. microscopy images or diffraction patterns) to interface with customers directly or extract inputs for simulations.
2. Stage 1 (8 teams, €1M per team): baseline technical benchmarks above and ability to interface with sample delivery solutions (below) will be demonstrated on the performer's

model system of choice. Model systems may make the demonstration easier than scalable industrial ones, e.g., gold nanoparticles have been a favorite model system for optical, x-ray, and electron tools [28–33] because of their high atomic number and refractive index.

3. Stage 2 (≤ 6 teams, €0.5M/team): performers will produce open data characterizing a diverse set of model yet industrially relevant materials. This data will be made public to foster simulation efforts by the community and to demonstrate the performers' credibility to possible industrial users.
    a. The government will provide model materials for stage 2 across diverse verticals and manufacturing methods, such as e.g. out-of-patent molecules to be crystallized, MOFs to be extruded, and so on.
    b. The government will select additional technical and commercialization benchmarks for stage 3 based on feedback from industry. To progress to stage 3, participants must demonstrate a path towards meeting them and develop commercialization plans.
4. Stage 3 (≤ 3 teams, €1M/team): participants will further meet industry-informed technical metrics and commercialization benchmarks.
    a. Direct qualification and testing on materials from possible partners will be facilitated.
5. The final prize to winning performers will be an advance commitment to purchase their winning solutions at the end of Stage 3 and make them available as a user facility, to be hosted at national laboratories and similar institutions for continued open use (e.g. at Max Planck or Fraunhofer institutes in Germany, or Royce Institute in the UK), up to €2M total. The institute will need to commit staff time towards future user support to be chosen as a host.

Additional funding is necessary for sample delivery development and ensuring the data makes its way to modellers who could use it (see "ecosystem creation" below). Flash will accomplish this via a separate small-grant track (up to €2M) and convenings.

Including 10% contingency, the total challenge budget request is €20M.

Challenge success metrics

The challenge seeks to accelerate scale-up and improvement to new and mature materials by making data on their nucleation and phase transformations available. This means that the data collection should become inexpensive, and the equipment or data itself should become usable by manufacturers directly and by computational chemical engineers.

The challenge should de-risk investment in both the performers and in manufacturing more broadly and improve the competitiveness of advanced materials manufacturing in Europe. At the end of the challenge, the performers will have:

- Brought the technical capacity of their toolkits to be sufficient for serving as real-time probes of materials manufacturing.
- Demonstrated their utility to manufacturing partners through both open data (stage 2) and achieving additional industry-informed metrics (stage 3).
- Produced open datasets of key exemplar materials for modelling of manufacturing processes at all scales, from atoms to ton-sized reactors, and placed this data in the hands of the modellers to do so. The modelling itself, however, is out of scope.

This challenge will have been a success if:

- Flash creates the capability to measure nucleation pathways and rates for practical manufacturing of industrially relevant materials.
- Performers sell equipment and attract follow-on investment, either to/by partners or otherwise. In turn, industry users or partners make productive use of the information provided by the performers.
- Performers and partners adapt the same equipment towards materials that were not immediately used within the challenge itself.

- New avenues of industry-facing research arise from the open dataset produced in stage 2, independent of the performers. These could include anything from atomistic simulation (Appendix E), reactor simulation, novel process design, or novel reactor design.

Ecosystem creation and transition support

To achieve the goals of the challenge, we seek to make it easy to make measurements on diverse materials and manufacturing routes and to connect the equipment makers to customers and computational engineers.

- Sample delivery: the engineers who make characterization equipment rarely possess the expertise to create small-scale reactors optimal for in-situ measurements, much less for multiple manufacturing routes. To deliver diverse samples into the measurement tools themselves at conditions relevant to manufacturing, Flash includes a small-scale (order €20K-100K per reactor) "sample delivery" track to create such reactors for the performers. The miniaturized reactors to be built may involve, e.g., fluidics, extruders, or heating stages, and the participants may range from traditional equipment companies to academics. Creating this track separately rather than relying on subcontracting enables interoperability across participants and avoids supplier lock-in. Success for sample delivery includes open-sourcing or commercializing the reactor solutions and verifying their compatibility with the measurement tools of main challenge participants. This will take place during stages 1-2.
- Connection to customers: the challenge helps the participants to reach sufficiently high TRLs to get their products directly into the hands of customers for qualification. The challenge will facilitate the connections to industrial customers via open days and open data on model materials. To simplify and accelerate agreements and contracts in stage 3 and beyond, the challenge team will draft standard material transfer agreements or relevant contracts.
- Building models: there are two communities of modellers, first-principles and fluid-dynamics. The first are at the focus of Appendix E for a prediction competition that is enabled by the challenge's data but is too early-stage to be funded by SPRIND. The second will be included in the jury to ensure challenge participants are producing usable and useful data. Flash seeks to engage process engineers from industry and academia as mentors in the challenge as well as to invite them to program open days and open participant presentations to ensure they are aware of the experimental capabilities being developed.

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These are the people with whom I have met or had calls to discuss the nucleation instrumentation challenge. They may be potential jury members or mentors, manufacturing partners, participants, transition partners, or people who make and sell materials for a living. I am especially grateful to my mentor Anne Fischer and the Renaissance Philanthropy Big if True Science team (Jean-Paul Chretien, Erin Szulman, and Iman Mouloudi).

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Alex Hammer, Dunia Innovations
Alex Chan, Ursa Labs
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Roham Dorakhan, Acceleration Consortium
Artem Mishchenko, Google DeepMind

Jehad Abed, Google X
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Will Richardson, Faraday Institution
Peter Abowd, Temasek
Coco Newton, Empirical Ventures
Chris Wilmer, Penn State, NUMat, Project URN
James Dillard, Macro Oceans
Ryan Chaban, ARPA-E and XPRIZE
Mike Fiddy, UNC Charlotte
Felix Lam, Brains fellow
Mervin Zhao, ARPA-E
Jay Lewis, Convergent FRO
Stephen Harris, Lawrence Berkeley Nat'l Lab
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